

Important Statistics Related to Order Statistics

- range of a random sample: $Y_n - Y_1$.
- midrange of a random sample: $(Y_1 + Y_n)/2$.

- Let Y_1, Y_2, Y_3 be the order statistics from a distribution with pdf $f(x) = 1, 0 < x < 1$. We seek the pdf of the range $Z = Y_3 - Y_1$. First

$$g_{13}(y_1, y_3) = 6(y_3 - y_1), \quad 0 < y_1 < y_3 < 1.$$

Let $Z_1 = Y_3 - Y_2$ and $Z_2 = Y_3$. The inverse is $y_1 = z_2 - z_1$ and $y_3 = z_2$. The Jacobian is $J = -1$. So the joint pdf of Z_1 and Z_2 is

$$h(z_1, z_2) = 6z_1 \quad 0 < z_1 < z_2 < 1.$$

Hence, the pdf of Z_1 is

$$h_1(z_1) = 6z_1(1 - z_1), \quad 0 < z_1 < 1.$$

- **Median:** $\xi_{0.5} = F^{-1}(\frac{1}{2})$
- Median for $N(1, 4)$: $\xi_{0.5} = 1$.
- **Sample Median:** If we observe random samples: $X_1, \dots, X_n$, let $Y_1 < Y_2 < \dots < Y_n$ be the order statistics, when n is odd, $Q_m = Y_{(n+1)/2}$, when n is even, $Q_m = (Y_{n/2} + Y_{n/2+1})/2$.
- Suppose we observe random sample 1, 6, 3, 7, 2, then median is 3. Suppose we observe 1, 6, 3, 7, median is $(6 + 3)/2 = 4.5$.

- Let X be a random variable with cdf $F(x)$. for $0 < p < 1$, define **p th quantile** of X to be $\xi_p = F^{-1}(p)$. e.g., $\xi_{0.5}$ is median.
- Let $Y_1 < \dots < Y_n$ be the order statistics. Let $k = [p(n+1)]$, use Y_k to be the estimator of ξ_p . Why?

$$\begin{aligned}\mathbb{E}(F(Y_k)) &= \int_a^b F(y_k) g_k(y_k) dy_k = \int_0^1 \frac{n!}{(k-1)!(n-k)!} z^k (1-z)^{n-k} dz \\ &= \frac{k}{n+1}\end{aligned}$$

- On the average, there is $k/(n+1)$ of the total area to the left of Y_k and $p \doteq k/(n+1)$. We call Y_k the **p th sample quantile**.
- Sample quantiles are useful descriptive statistics. For example, if Y_k is the p th quantile of the sample, then we know approximately $p100\%$ of the data are less than or equal to Y_k . e.g., GRE test.

- A **five number** summary of the data: the minimum (Y_1), the first quartile ($Y_{[0.25(n+1)]}$), the median ($Y_{0.5(n+1)}$ odd sample), the third quartile ($Y_{[0.75(n+1)]}$), and the maximum (Y_n).
- **Boxplot** of the data.